NUS GRIP
Ambient Systems

Climate Management for Cities of Tomorrow

Assignment #6
October 27, 2020

Introduction

Vision: create a safe, healthy, and productive indoor environment for everyone.

Mission: become a leading global provider of comprehensive smart building solutions.

Goal: develop new products enabled by IoT and that delight property owners and building operators, starting with the indoor farming segment.



Overview

1. Problem
2. Solution
3. Market
4. Business Model
5. Competitive Analysis
6. Financials
7. Milestones
8. Traction
9. Team

Section 1: Problem



Problem Statement



Climate Change

Need for greater control over indoor environments



Urban Pollution and Indoor Contaminants

Need for cleaner air



Rising Energy Costs

Need to keep energy consumption low



Increasing Technological Complexity

Need for simple solutions

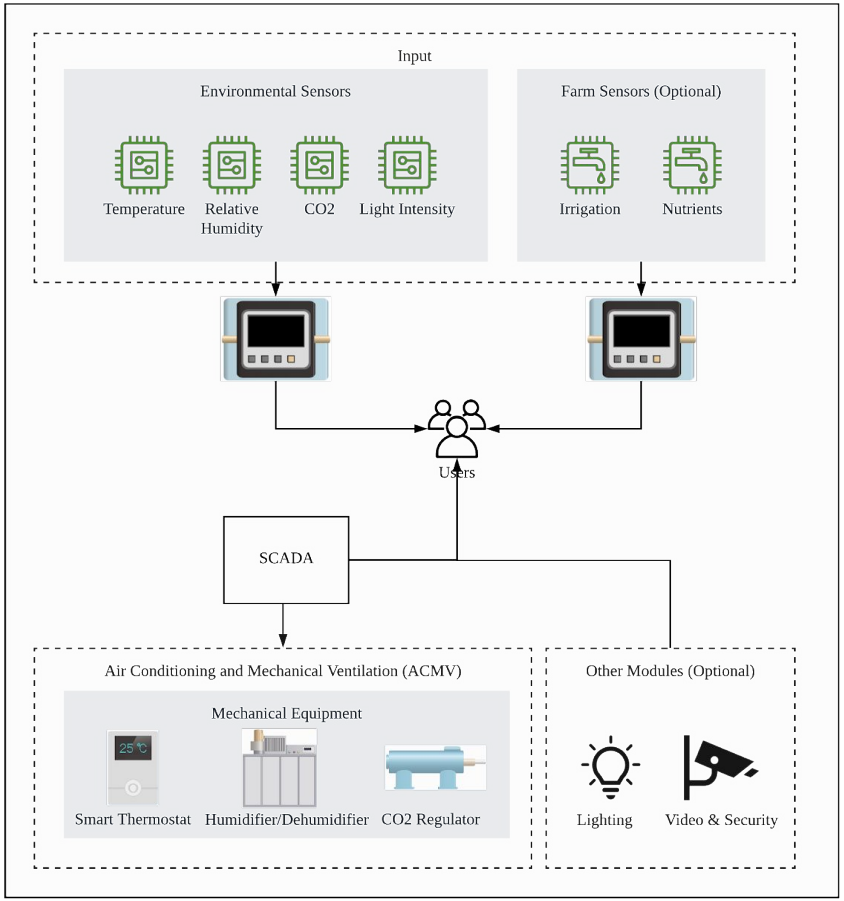
Top Five Pain Points of Property Owners

Top **five pain points** of property owners and building operators across building types:

1. Hard to find trained personnel to continuously manage building's indoor environment
2. No or minimal visibility into real-time environmental parameters
3. Outdated and incompatible legacy systems
4. Competing priorities (e.g., improve environmental parameters or reduce energy costs)
5. Evolving legal compliance and certification requirements

Key takeaway: building operators face similar challenges across different building types.

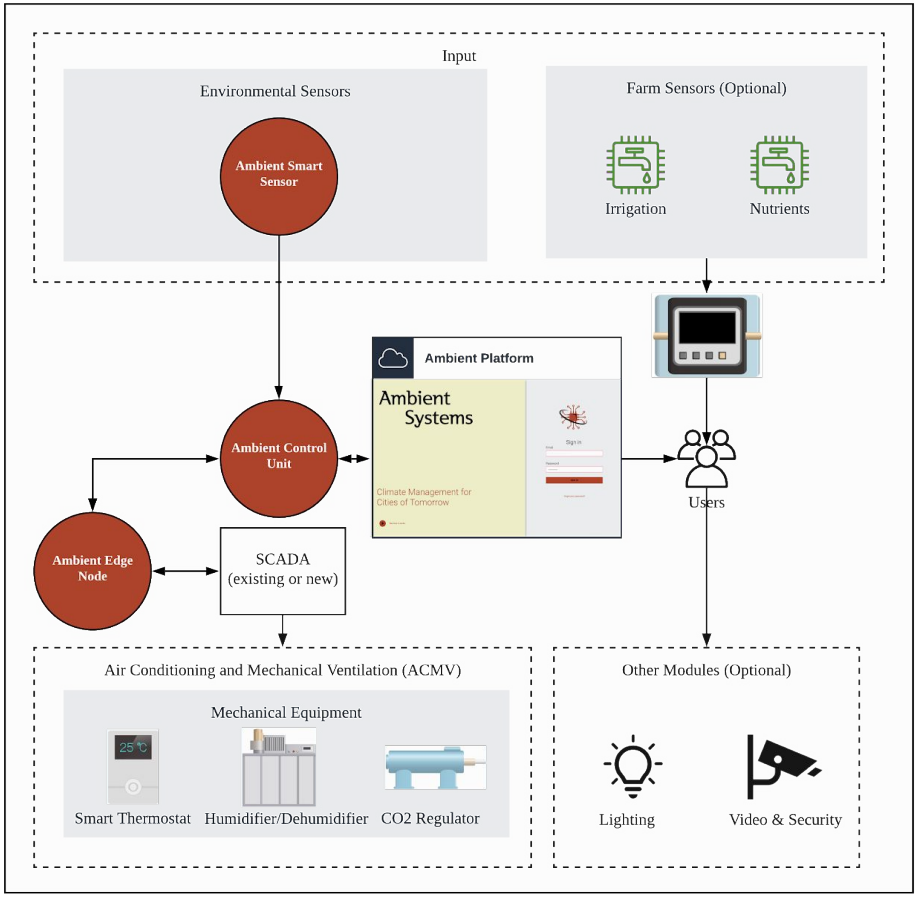
Current Scenario



Section 2: Solution



Ambient Solution



Technology Stack

Ambient Smart Sensors	Ambient Edge Node	Ambient Central Control Unit	Ambient Platform
• Four (4) core parameters: Temperature, Relative Humidity, CO2, Light • Optional parameters: VoC, Occupancy, Sound, Pressure, and others	• Software-Defined Hardware adapts and controls a variety equipment • Control extensions: Dampers, VAV, VFD, Lighting, Mixing Valves, Pumps • Built-in programmable logic controller (PLC) • Connected to primary Air Handling Units (AHUs) or Air Conditioners (AC)	• Manages communication between sensors, edge nodes, and • Functions as gateway between IoT network and the internet • Provides functionality at the network edge	• Cloud-based dashboard provides data access and analytics • Includes five (5) modules: facility analytics, ACMV optimization, maintenance, farm management, and marketplace • Sends alerts to users with recommendations for automatic improvements

Ambient Systems

Climate Management for Cities of Tomorrow

See how it works



Sign in

Email

Password

SIGN IN

Forgot your password?

Ambient Sytems

Username



Farm #1

38 College Ave, Singapore

- Overview
- Yield Prediction
- Analytics
- ACMV Optimisation
- Maintenance
- Marketplace
- Settings

Overview

Camera



Yield Prediction

Crop Yield

Yield Week

13

1.7717 kg/m2

Crop Quality

Yield Week

13

1.7717 kg/m2

Anticipated Harvest Time

December 2020

Mo	Tu	We	Th	Fr	Sa	Su
1	2	3	4	5	6	7
8	9	10	11	12	13	14
15	16	17	18	19	20	21
22	23	24	25	26	27	28
29	30	1	2	3	4	5

Climatic Conditions

TEMPERATURE	RELATIVE HUMIDITY	CO2	LIGHT INTENSITY
20.5	55%	1050	600

Energy Consumption



AC	16%
Humidifier	22%
CO2 Regulator	11%
Ventilation	39%
Lighting	12%

Section 3: Market



Total Available Market (TAM)

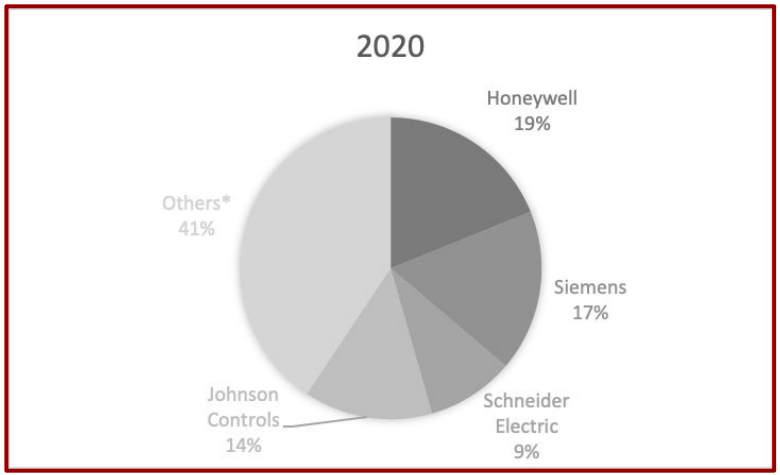
$4.74B

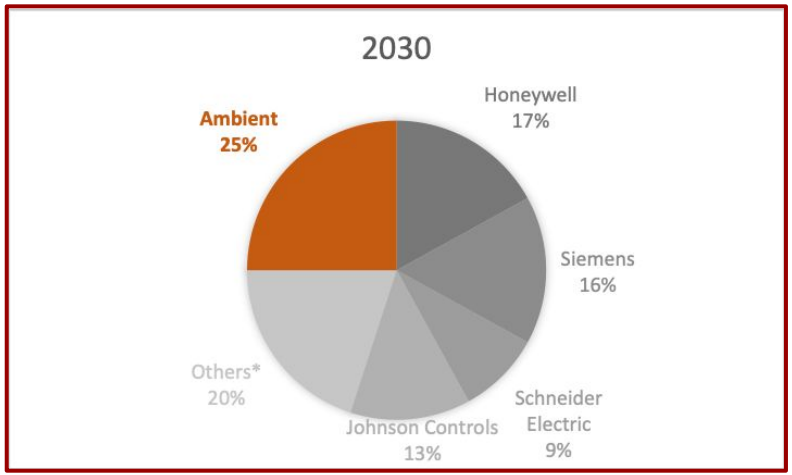
Total BAS Market: Revenue Forecast by Region, Global, 2017–2026

Year	Europe ($ Million)	North America ($ Million)	APAC ($ Million)	Latin America ($ Million)	MEA ($ Million)
2017	1,776.0	1,232.9	1,107.2	106.9	197.2
2018	1,823.5	1,269.9	1,149.2	111.0	203.5
2019	1,865.5	1,310.5	1,195.2	115.6	210.0
2020	1,878.5	1,321.3	1,210.2	117.7	212.9
2021	1,927.9	1,373.3	1,282.1	125.8	225.2
2022	1,963.6	1,420.8	1,340.7	131.5	234.1
2023	2,007.4	1,472.3	1,408.1	137.0	242.8
2024	2,059.8	1,523.4	1,483.4	143.2	253.3
2025	2,121.4	1,584.8	1,560.5	149.4	263.5
2026	2,184.8	1,655.2	1,646.5	155.4	274.9
CAGR	**2.3%**	**3.4%**	**4.7%**	**4.3%**	**3.9%**

Data from Frost & Sullivan

Serviceable Obtainable Market (SOM)

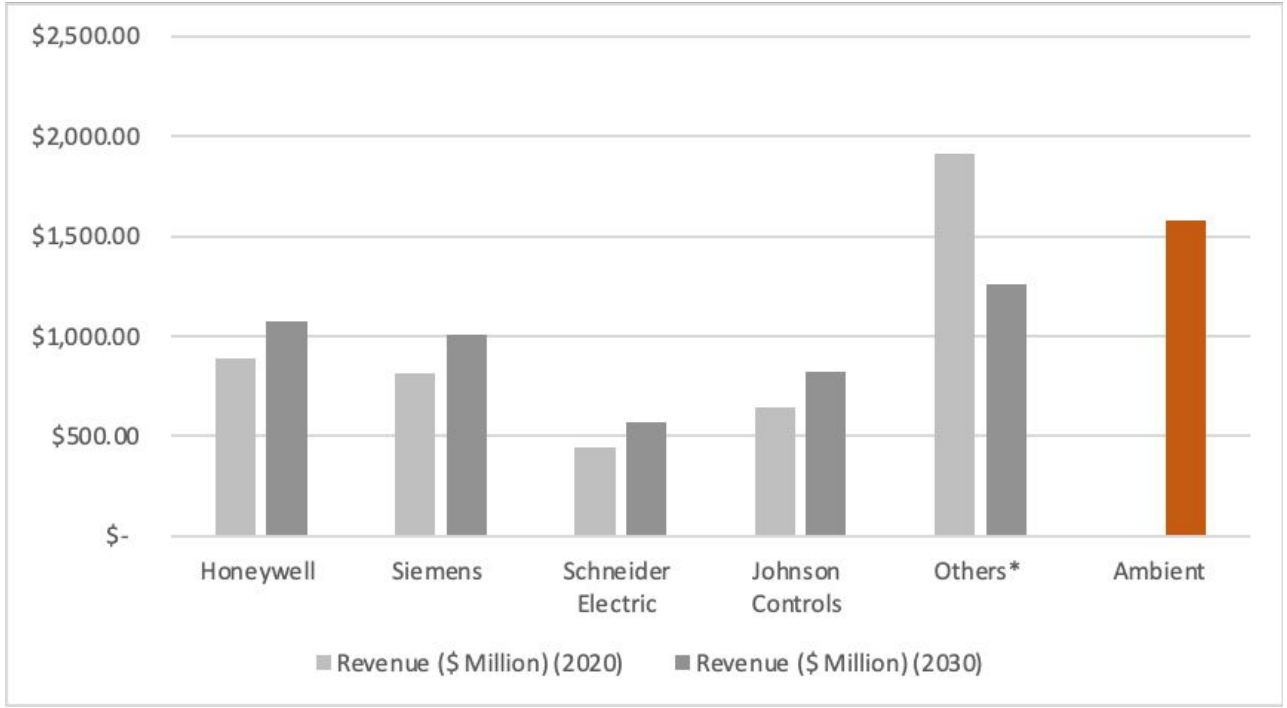




Data from Frost & Sullivan

Serviceable Obtainable Market (SOM)

$1.57B



Data from Frost & Sullivan

Bowling Pin Diagram

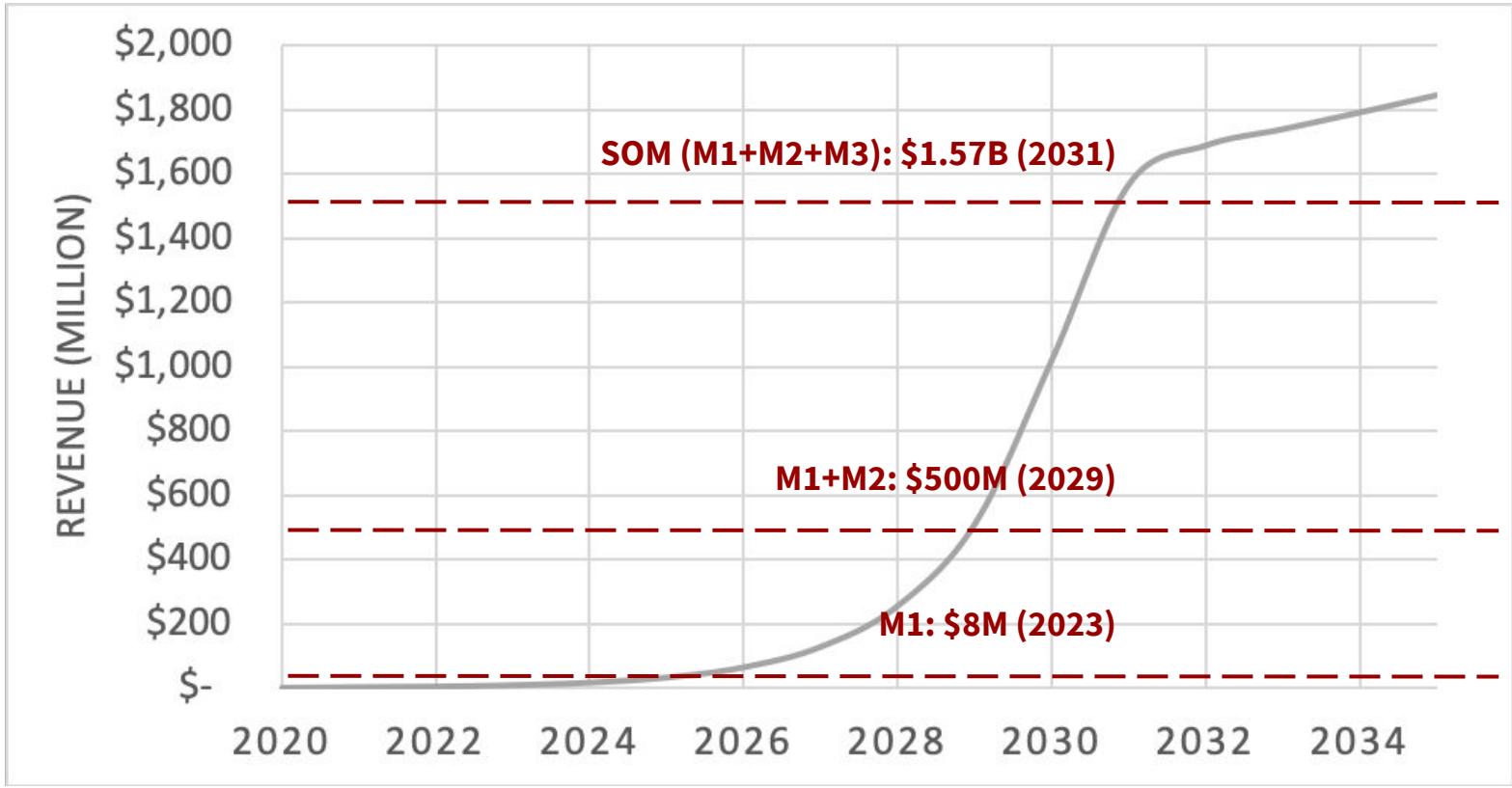
Market Legend

1. M1: SG Indoor F
2. M2: APAC C&I (5
3. M3: Europe C&I (50% by 2031)

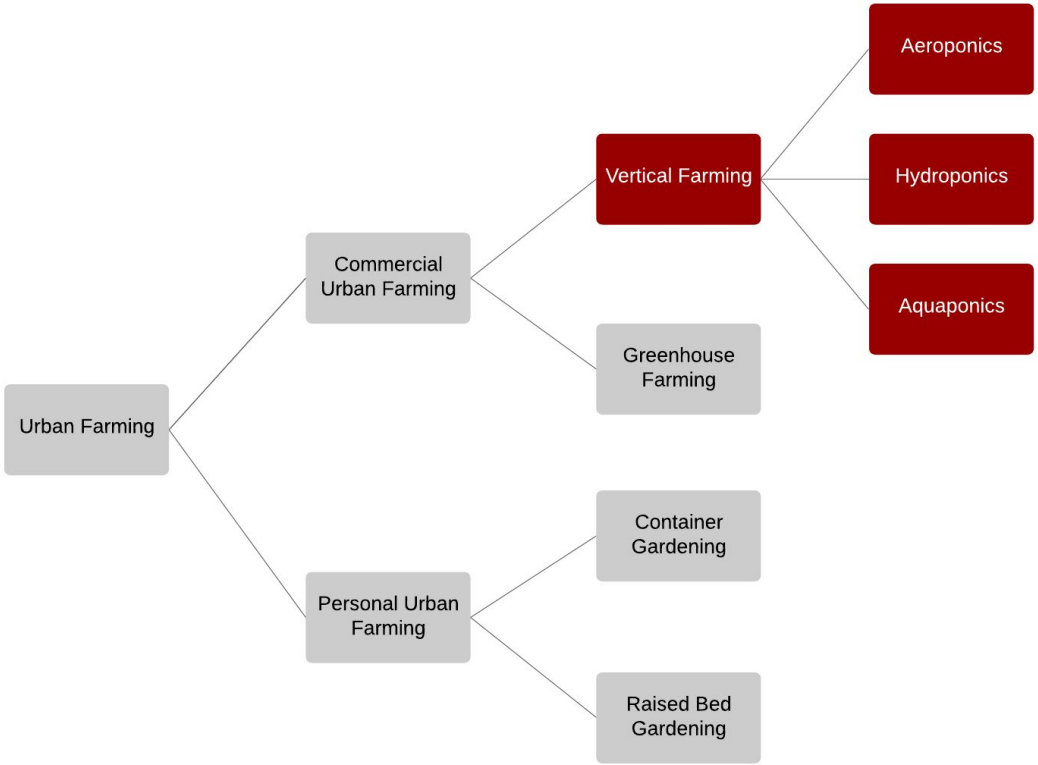
e Climate Management

3.

3s Framework

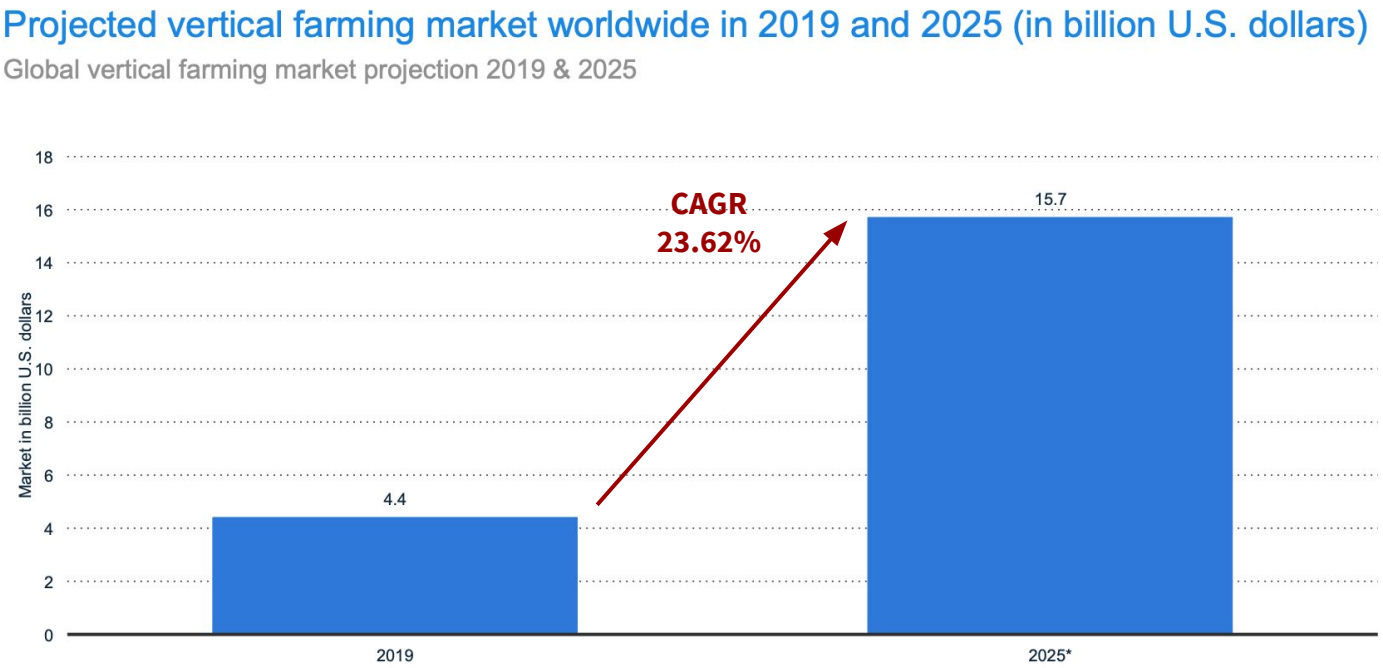


Urban Farming Market Segmentation



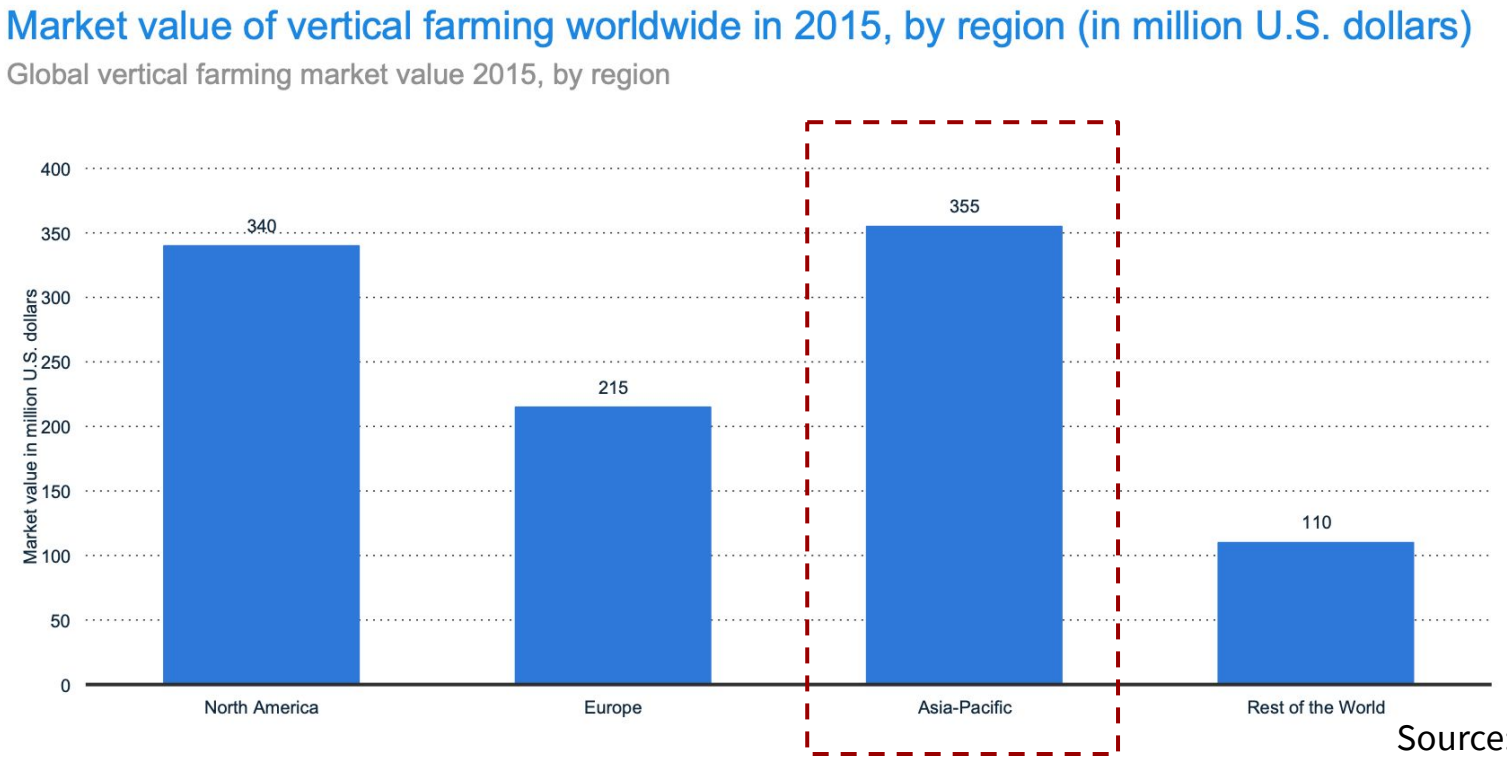
Source: Frost & Sullivan

Vertical Farming Market



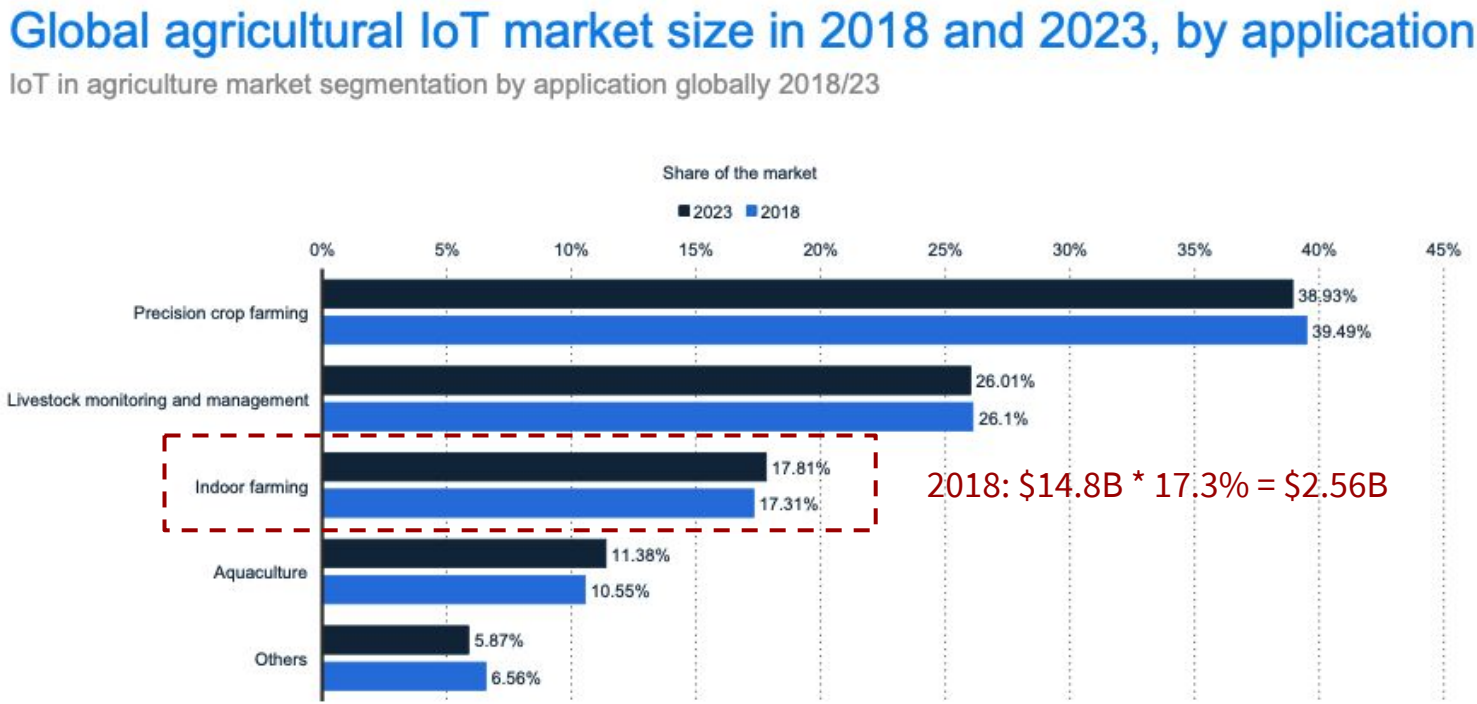
Source: Statista

Vertical Farming Market



Source: Statista

Agricultural IoT Market



2018: $14.8B * 17.3% = $2.56B

Source: Statista

Why start with Indoor Farms?

1. Vertical Farming is a new and rapidly growing industry.

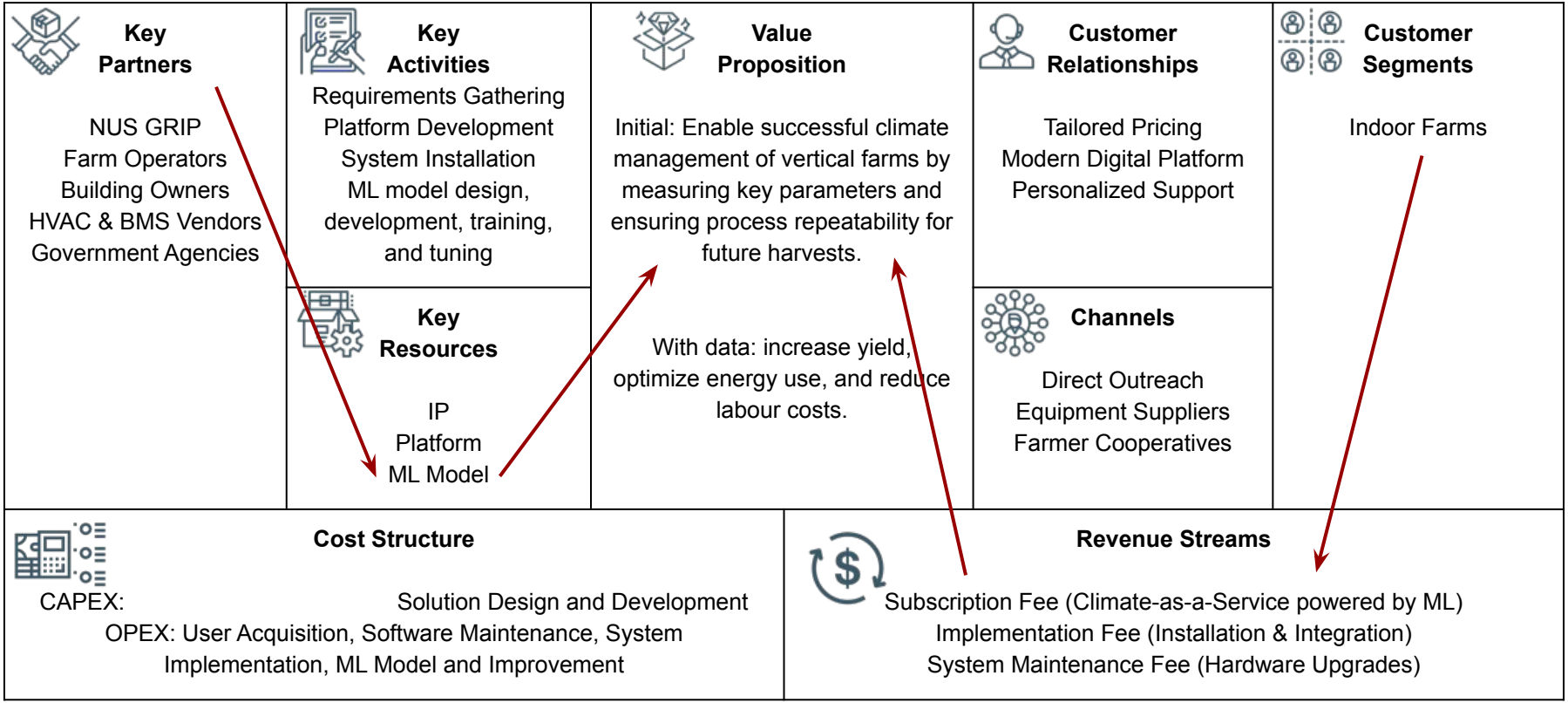
2. The potential market that can be addressed is greater than $1B.

3. Farm operators have indicated interest and need for this type of solution.

Section 4: Business Model

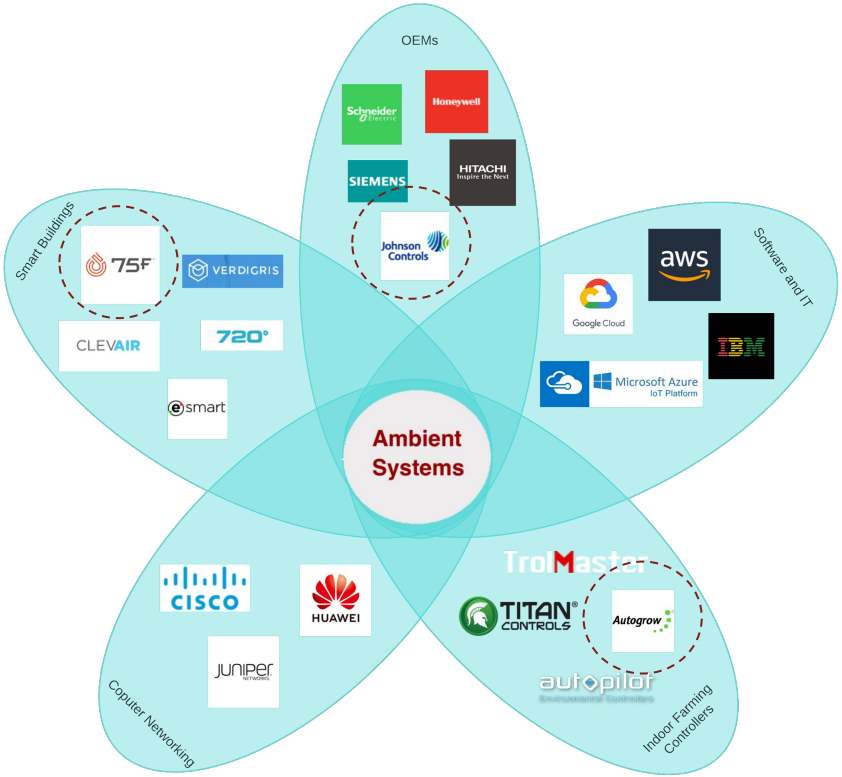


Business Model Canvas



Section 5: Competitive Analysis

Competitive Analysis: Petal Diagram



Ambient is positioned at the intersection of multiple segments in order to provide a solution tailored for the indoor farming market.

Competitive Benchmark

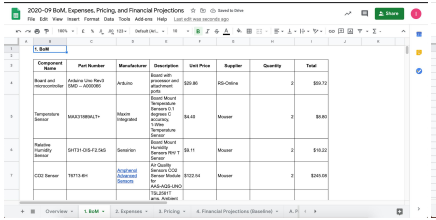
	Ambient	Autogrow	75F	
Segment	Multiple	Indoor Farming Controller	Smart Buildings	OEM
Proximity to Customer	✓ Singapore	✗ US	✗ US	✓ International
Specific vs. Generic	✓ Indoor Farms	✓ Indoor Farms	✗ No	✗ No
Modern Web-based UI	✓ Yes	✗ No	✓ Yes	✗ Some
Wireless IoT Network	✓ Yes	✗ No	✓ Yes	✓ Yes
Advanced Optimization Methods	✓ ML models	✗ No	✗ Some	✗ Some
Pricing	Quotation (based on area)	$3000 for one unit + Installation Fees	Quotation	Quotation

Section 6: Financial Projections

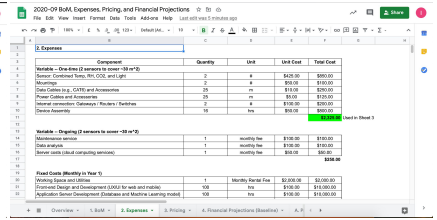


Financial Model Process

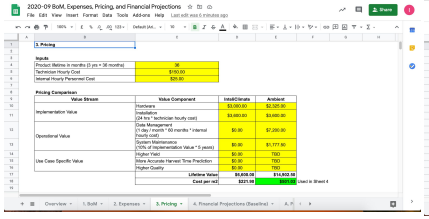
1. BoM



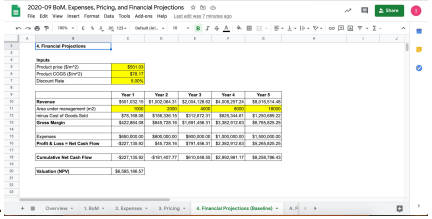
2. Expenses



3. Pricing



4. Financial Projections



Hardware Costs

Variable Expense

COGS and Price

NPV and Funding Gap

Section	Component	Cost
CAPEX		
Services	Audit and Design	Complimentary
	Installation	$4,000 - $5,000
Hardware	Ambient Smart Sensors	$1,500 - $2,500
	Ambient Edge Node	$2,500 - $4,500
	Ambient Control Unit	$4,000 - $6,000
	CAPEX SubTotal	**$12,000 - $18,000**
OPEX		
Operations	Secure Data Management	Complimentary
	System Maintenance	Complimentary
	Parts Replacement	Complimentary
Ambient Platform	Facility Analytics	Value-sharing
	ACMV Optimization	Value-sharing
	Predictive Maintenance	Value-sharing
	Yield Prediction	Value-sharing
	Marketplace	Value-sharing
	OPEX SubTotal	**$0**
	Estimated Total Cost	**$12,000 - $18,000 + OPEX value-sharing**

Financial Projections: Baseline

Inputs

Product price ($/m^2)	$500.00
Product COGS ($/m^2)	$80.00
Discount Rate	5.00%

	Year 1	Year 2	Year 3	Year 4	Year 5
Revenue	$500,000.00	$1,000,000.00	$2,000,000.00	$4,000,000.00	$8,000,000.00
Area under management (m2)	1000	2000	4000	8000	16000
minus Cost of Goods Sold	$80,000.00	$160,000.00	$320,000.00	$640,000.00	$1,280,000.00
Gross Margin	$420,000.00	$840,000.00	$1,680,000.00	$3,360,000.00	$6,720,000.00
Expenses	$650,000.00	$800,000.00	$900,000.00	$1,000,000.00	$1,500,000.00
Profit & Loss = Net Cash Flow	-$230,000.00	$40,000.00	$780,000.00	$2,360,000.00	$5,220,000.00

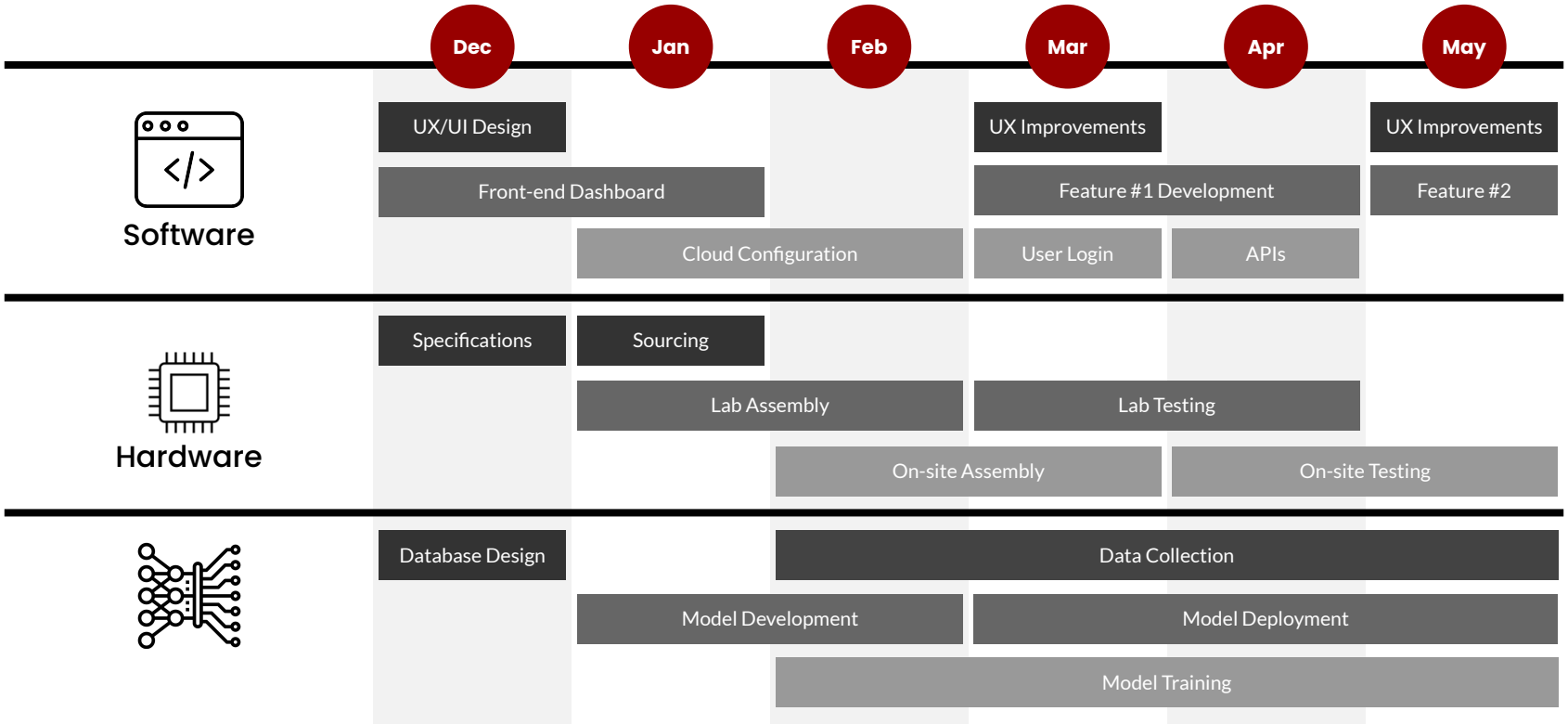
	-$230,000.00	-$190,000.00	$590,000.00	$2,950,000.00	$8,170,000.00

Valuation (NPV)	$6,522,611.32

Section 7: Milestones



PRODUCT ROADMAP



Section 8: Traction

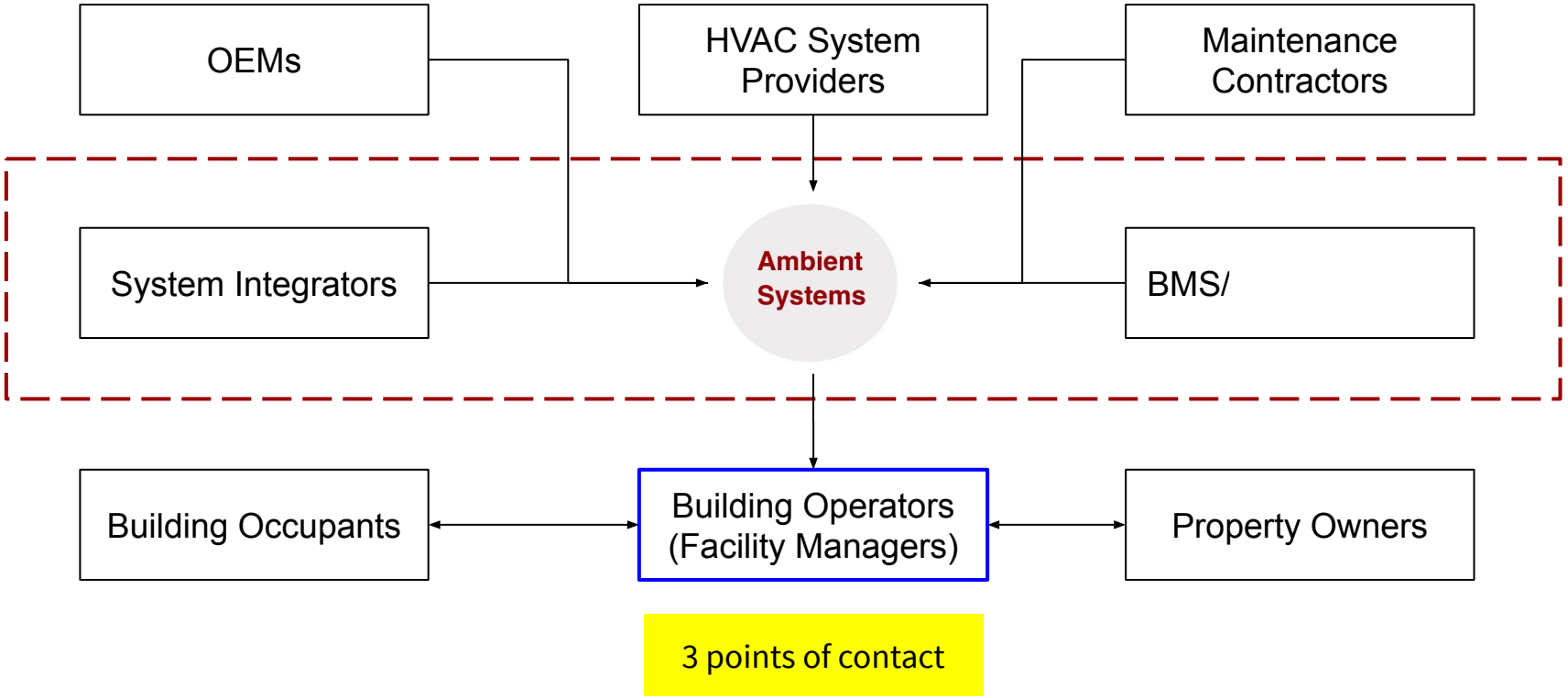


Section 9: Team

Thank you

Appendix

Value Chain Analysis



MVP: Two Paths

Path	Original Equipment Manufacturer (OEM)	System Integrator
Activities	1. Design IoT device based on 2. Procure components 3. Assemble device with casing 4. Develop firmware 5. Install on-site	1. Procure controller and sensors 2. Install on-site 3. Connect with existing systems
Advantages	• Flexible customization • Financially sustainable	• Ready to use • Manufacturer support
Disadvantages	• Electronics design cost and complexity • Manufacturing and distribution costs	• Solutions are wired • No data analytics capabilities • Manufacturers are far away


Device Enclosure


Temp sensor

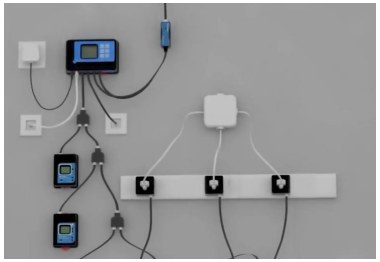

CO2 sensor


Micro-controller (“the platform”)


RH sensor



Transceiver


Connected controller and sensors


Controller


Sensors

Bill of Materials: IoT Sensors

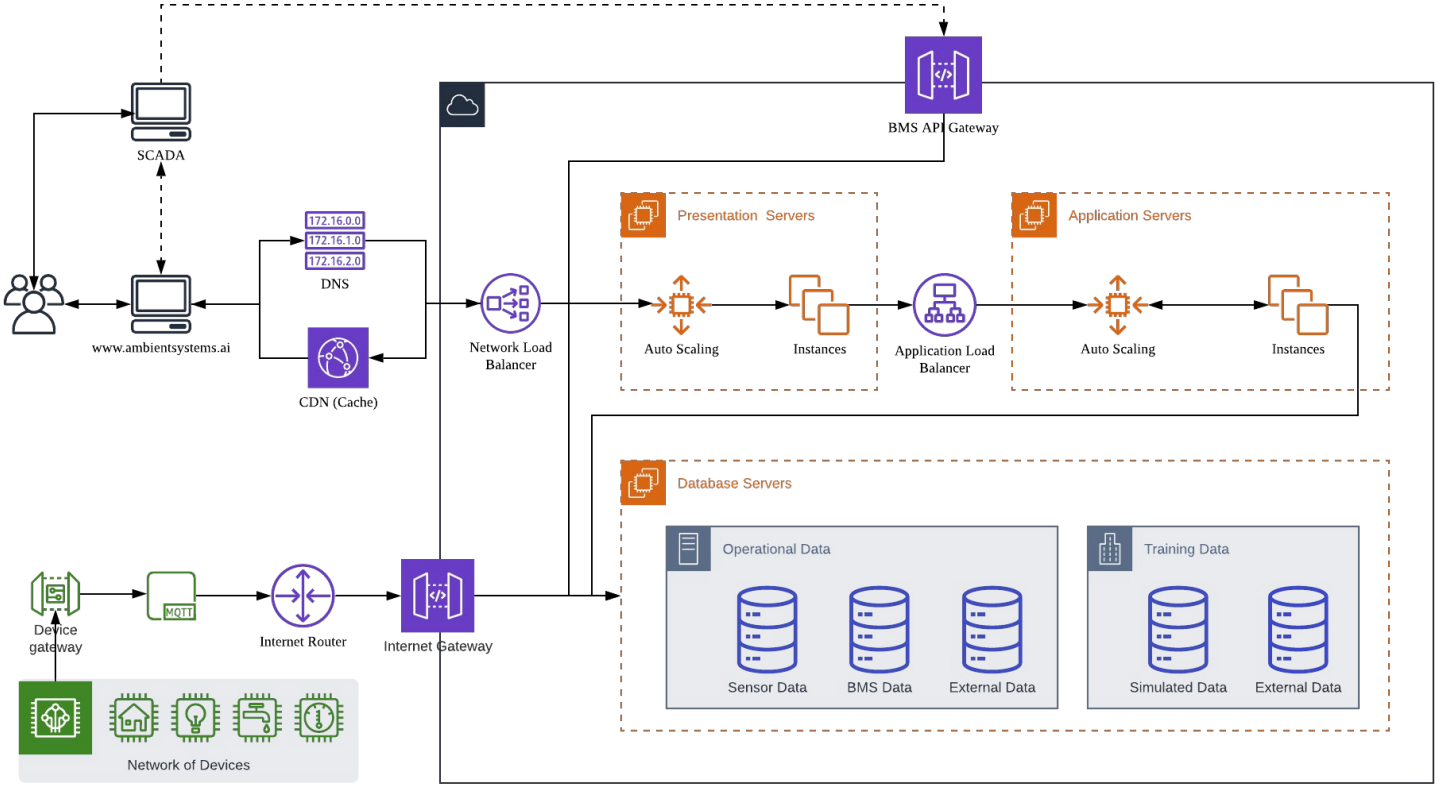
Component Name	Part Number	Manufacturer	Description	Picture	Unit Price	Supplier	Quantity	Total
Board and microcontroller	Arduino Uno Rev3 SMD -- A000066	Arduino	Board with processor and attachment ports		$29.86	RS-Online	2	$59.72
Temperature Sensor	MAX31889ALT+	Maxim Integrated	Board Mount Temperature Sensors 0.1 degrees C accuracy, 1-Wire Temperature Sensor		$4.40	Mouser	2	$8.80
Relative Humidity Sensor	SHT31-DIS-F2.5kS	Sensirion	Board Mount Humidity Sensors RH/ T Sensor		$9.11	Mouser	2	$18.22
CO2 Sensor	T6713-6H	Amphenol Advanced Sensors	Air Quality Sensors CO2 Sensor Module for AAS-AQS-UNO		$122.54	Mouser	2	$245.08
Light Sensor	TSL2561T ams	ams	TSL2561T ams, Ambient Light Sensor Unit Surface Mount 6-Pin TMB		$4.41	RS-Online	2	$8.82
Transceiver	LR1110IMLTRT	Semtech	RF Transceiver LONG RANGE WI-FI LOC LORA TRANSCEIVER		$14.93	Mouser	2	$29.86
ADCs	AD7795BRUZ	Analog Devices	Analog Devices AD7795BRUZ, 16-Bit Serial ADC Differential Input, 24-Pin TSSOP		$16.02	RS-Online	8	$128.16
Device enclosure	NSYTBP342912T	Schneider Electric	Plastic Enclosure, Flame Retardant, Multipurpose, Polycarbonate, 341 mm, 291 mm, 128 mm, IP66		$106.51	Element 14	2	$213.02
							Total	**$720 for 2 sensors**

Detailed bill of materials for the OEM MVP path has been been made for IoT Sensors

Standards and Certifications

Singapore	Europe and US	International
IMDA: ● TR 47 : 2016 IoT reference architecture for Smart Nation ● TR 50 : 2016 IoT information and services interoperability for Smart Nation ● IMDA IoT Cyber Security Guide (2019) ● MTCS Certification Scheme	CISPR 22 standard for Information Technology Equipment-Radio Disturbance Characteristics-Limits and Methods of Measurement. More specifically: ● EN55022 ● EN61000-4-2 ● EN61000-4-3 ● EN55024 CE marking Federal Communications Commission (FCC) ● Part 15 Class A	ISO/IEC: ● 90003:2014: Software engineering ● 27000:2018: Information technology — Security techniques — Information security management systems

Preliminary System Architecture



Previous Pricing: TEV and PEV

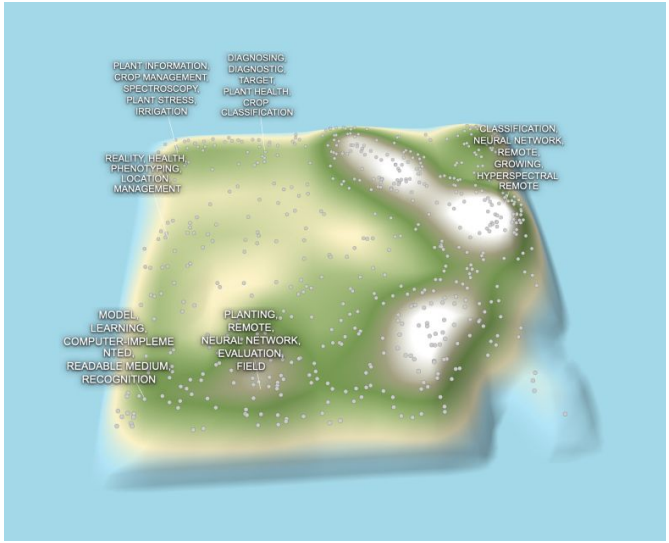
Value Stream	Value Component*	Best Alternative	Ambient
Implementation Value	Sensing Hardware	$3,000.00	$2,325.00
	Installation (24 hours * $150/hr technician cost)	$3,600.00	$3,600.00
Operational Value	Data Management (1 day / month * 36 months product lifetime * $25/hr personnel cost)	$0.00	$7,200.00
	System Maintenance (10% of Implementation Value * 5 years)	$0.00	$1,777.50
Use Case Specific Value	Higher Yield	$0.00	TBD
	More Accurate Harvest Time Prediction	$0.00	TBD
	Higher Quality	$0.00	TBD
	Total Economic Value (TEV)	**$6,600.00**	**$14,902.50**
	TEV per m2 of farming area	**$221.90**	**$501.03**

*Main Assumptions: product lifetime and subscription contract of 3 years (36 months); 1 farming unit covers 30m2.

MVP: Model

- System lays down the foundation for the prediction and optimization model
- Numerous options (depends on requirements)
- Potential: neural network (supervised learning)
- Consists of several “layers” of data -- potential structure:
  - Layer one: sensor measurements (i.e., raw sensor data)
  - Layer two: equipment operational logs
  - Layer three: output (yield, harvest time, crop quality)
- Use-cases:
  - Receive recommendations for climate controls (e.g., adjust temperature, increase humidity)
  - Predict crop output, harvest time, and quality
  - Automate climate controls

Draft: Patent Landscape





Prior Art Search

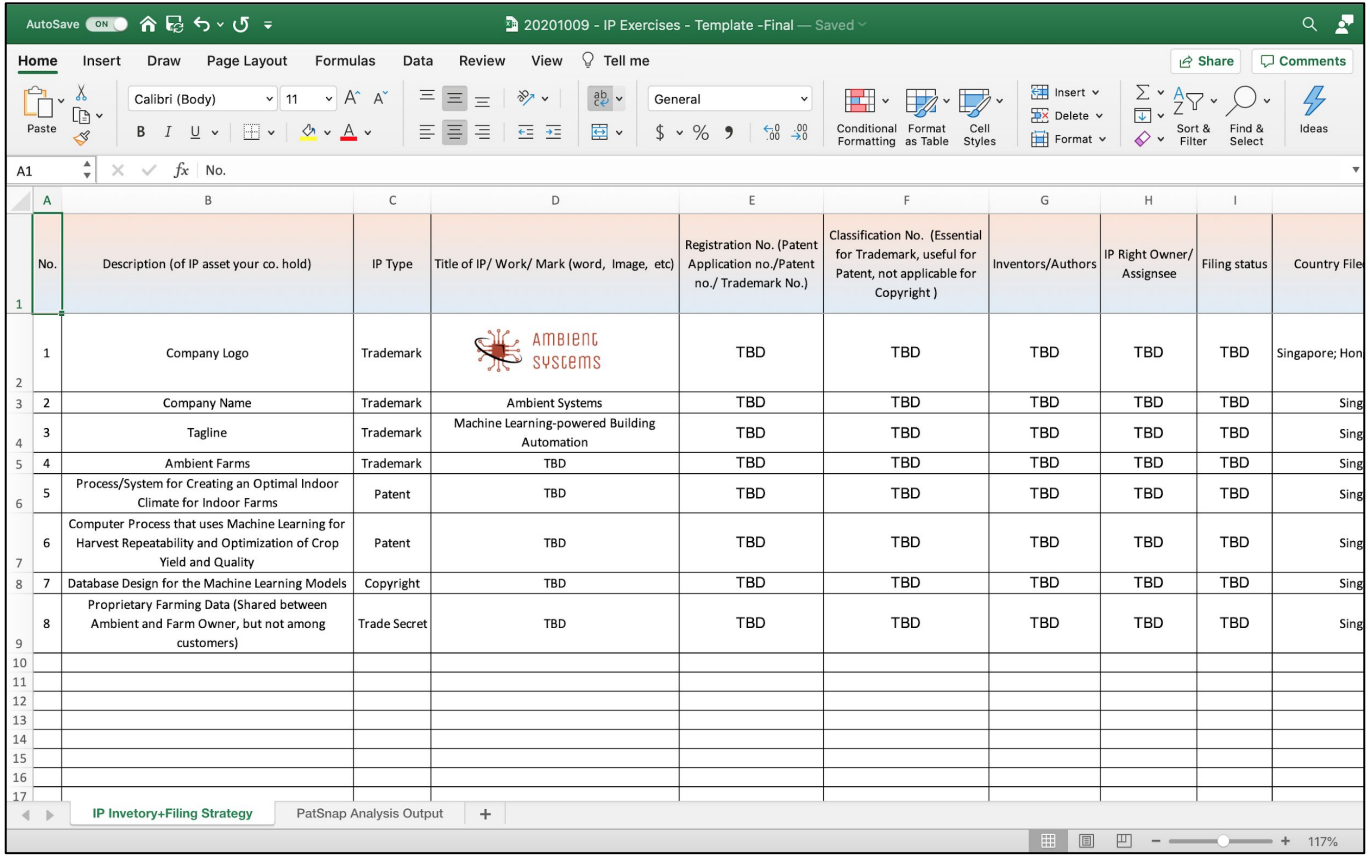
Smart Buildings: HVAC Control Patents	AgTech: Indoor Farming Patents
● US 10,798,177 B2: HVAC Controller with Integrated Wireless Network Processor Chip, 2020 ● US 9,429,923: HVAC Control System and Method, 2016	● US 10,803,312 B2: AI-Powered Autonomous Plant-Growth Optimization System that Automatically Adjusts Input Variables to Yield Desired Harvest Traits, 2020 ● WO 2020/167934 A1: Controlled Agricultural Systems and Methods of Managing Agricultural Systems, 2020

Patent Analysis

Building Automation System (BAS) + Indoor Farm Optimization and Forecasting

Part 1: Building Automation System (BAS)	Part 2: Indoor Farm Optimization and Forecasting
● Uses an IoT network of sensors to collect data on building conditions, namely temperature, relative humidity, CO2 levels, and light intensity (for facilities with natural lighting) ● Provides hardware attachments for data collection and control of the existing SCADA system ● Provides real-time snapshot of environmental conditions at the facility, segmented per room ● Determines equipment set-points: thermostat, humidifier / de-humidifier, fans for ventilation and related equipment ● Provides automatic control for mechanical components that influence indoor climate (e.g., natural ventilation and lighting using windows)	● Combines data from IoT network of sensors and farm operations to determine optimal growing conditions ● Ensures farming process repeatability using a proprietary database design and supporting data warehousing services ● Uses a supervised neural network (i.e., a method) to optimize for crop yield and quality ● Improves accuracy of harvest time forecasts by determining correlation between harvest timing and related climate and non-climate parameters ● Uses multiple harvest cycles to train the model (each additional cycle enhances model effectiveness and accuracy)

IP Inventory/IP Filing Strategy



AutoSave ON — 20201009 - IP Exercises - Template -Final — Saved

Home Insert Draw Page Layout Formulas Data Review View Tell me — Share — Comments

Calibri (Body) 11 — General — Conditional Formatting — Format as Table — Cell Styles — Insert — Delete — Format — Sort & Filter — Find & Select — Ideas

A1 — No.

No.	Description (of IP asset your co. hold)	IP Type	Title of IP/ Work/ Mark (word, Image, etc)	Registration No. (Patent Application no./Patent no./ Trademark No.)	Classification No. (Essential for Trademark, useful for Patent, not applicable for Copyright)	Inventors/Authors	IP Right Owner/ Assignsee	Filing status	Country File
1	Company Logo	Trademark	Ambient Systems	TBD	TBD	TBD	TBD	TBD	Singapore; Hon
2	Company Name	Trademark	Ambient Systems	TBD	TBD	TBD	TBD	TBD	Sing
3	Tagline	Trademark	Machine Learning-powered Building Automation	TBD	TBD	TBD	TBD	TBD	Sing
4	Ambient Farms	Trademark	TBD	TBD	TBD	TBD	TBD	TBD	Sing
5	Process/System for Creating an Optimal Indoor Climate for Indoor Farms	Patent	TBD	TBD	TBD	TBD	TBD	TBD	Sing
6	Computer Process that uses Machine Learning for Harvest Repeatability and Optimization of Crop Yield and Quality	Patent	TBD	TBD	TBD	TBD	TBD	TBD	Sing
7	Database Design for the Machine Learning Models	Copyright	TBD	TBD	TBD	TBD	TBD	TBD	Sing
8	Proprietary Farming Data (Shared between Ambient and Farm Owner, but not among customers)	Trade Secret	TBD	TBD	TBD	TBD	TBD	TBD	Sing

IP Invetory+Filing Strategy — PatSnap Analysis Output — 117%

Growth Strategy

Start with Building Automation System (BAS) for Indoor Farms

Two ways to cross the chasm:

1. Become an integrated Smart Building solution provider by moving into hardware manufacturing under the "Ambient" brand name
    - Provide IoT sensor design and assembly
    - Build an in-house hardware controller for HVAC management and automation
    - Create custom equipment automation components (e.g., actuators, dampers)
2. Move towards an Agtech Platform by leveraging network effects and more accurate ML forecasts
    - Bring in nutrient suppliers
    - Bring in technology equipment suppliers (e.g., lighting, racks, sensors)
    - Connect farmers with distributors or end buyers (e.g., restaurants, super markets, salad stores)
    - Connect farmers with insurers
    - Connect farmers to Food Processing companies

Key stakeholders

Potential Competitor(s):

-
- Environmental Controller Companies

Potential Collaborator(s):

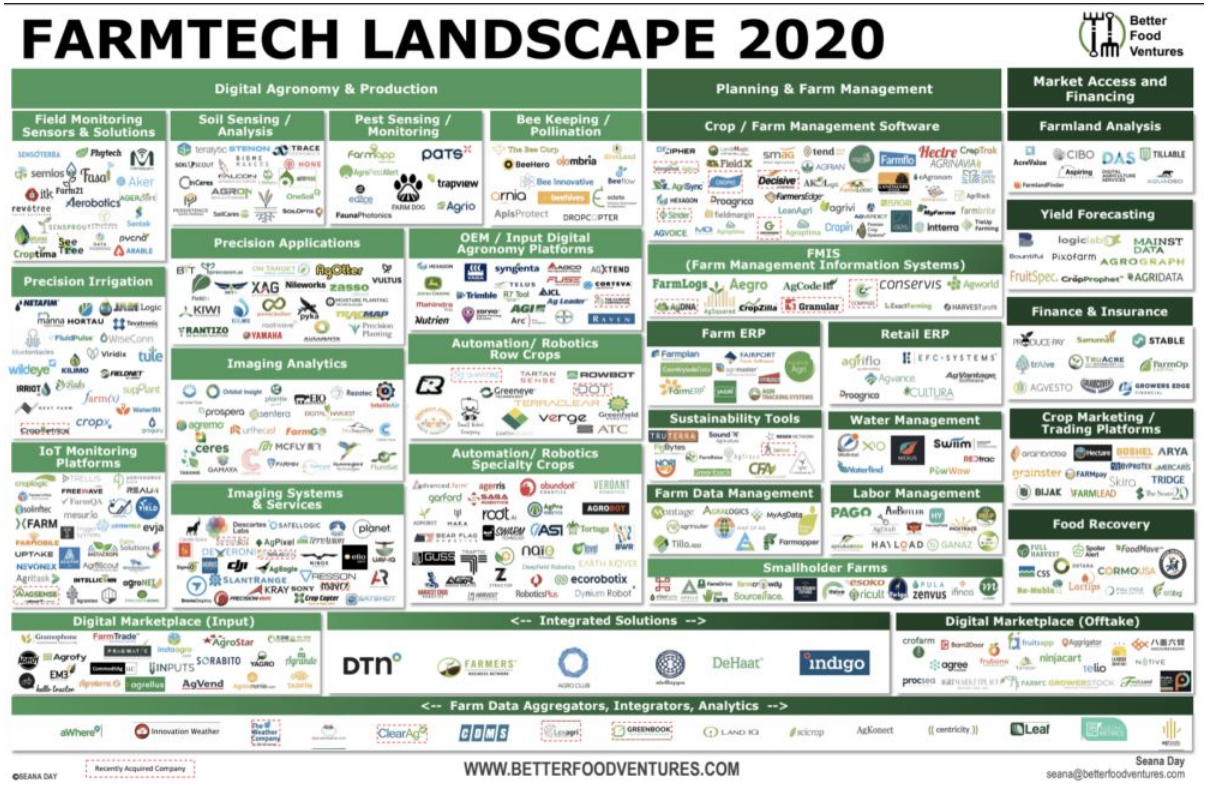
- OEMs: , Siemens, Honeywell,
- Smart Building Companies

Potential Infringer(s):

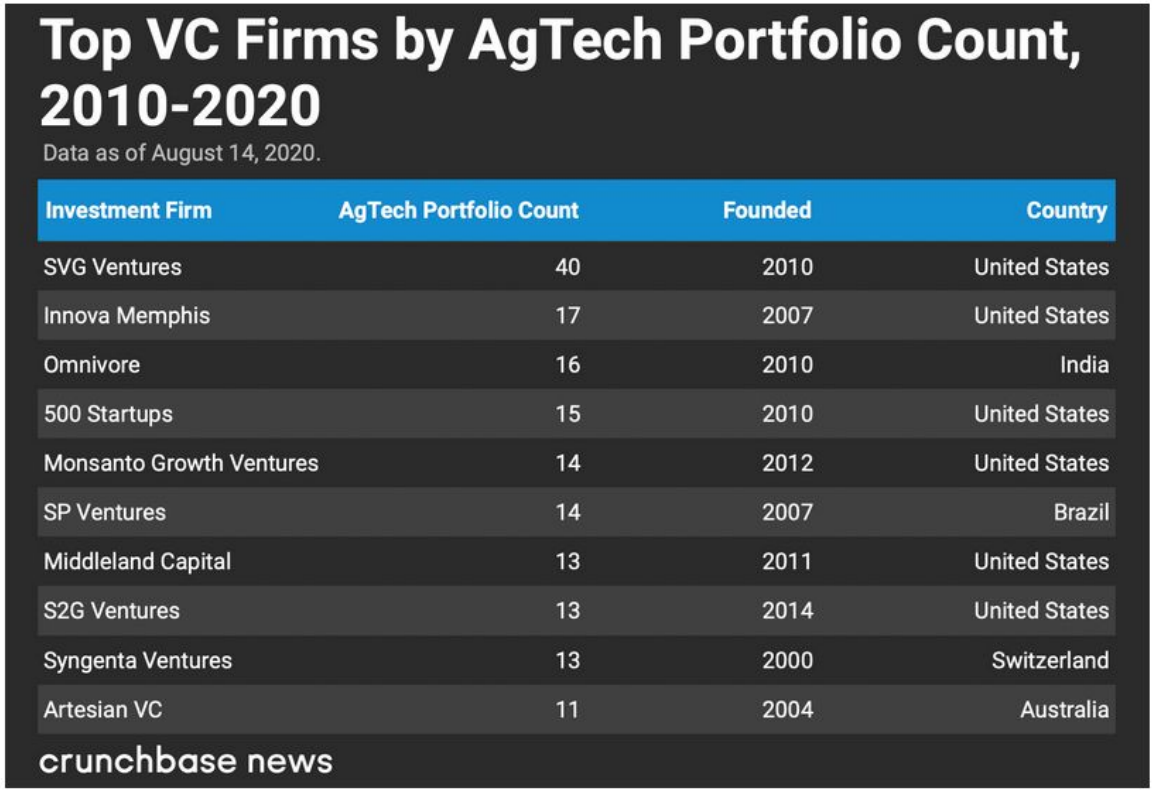
-
- Environmental Controller Companies

Potential Licensee(s):

- Indoor Farming Companies (i.e., farm operators): Plenty, Aerofarms



Potential Investors



Top VC Firms by AgTech Portfolio Count, 2010-2020

Data as of August 14, 2020.

Investment Firm	AgTech Portfolio Count	Founded	Country
SVG Ventures	40	2010	United States
Innova Memphis	17	2007	United States
Omnivore	16	2010	India
500 Startups	15	2010	United States
Monsanto Growth Ventures	14	2012	United States
SP Ventures	14	2007	Brazil
Middleland Capital	13	2011	United States
S2G Ventures	13	2014	United States
Syngenta Ventures	13	2000	Switzerland
Artesian VC	11	2004	Australia

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Accelerators

- 500 Startups
- The Yield Lab
- Iowa AgriTech Accelerator
- Y Combinator
- Techstars: Farm to Fork
- Better Food Ventures

Source: https://news.crunchbase.com/news/agtech-sector-blooms-as-more-dollars-and-startups-rush-in/